FITTING MACHINE LEARNING MODELS FOR DECIPHERING GRAFITTI

PRESENTED TO:
LAW ENFORCEMENT

PRESENTED BY:
JOHN STEELE

OVERVIEW

- PROPOSAL
- TECHNOLOGIES
- DATA SET AND CLEANING
- DATA TESTING
- ACCURACY
- RECAP

PROPOSAL

CAN WE FIT A MACHINE LEARNING MODEL TO BE ABLE TO READ GRAFITTI?

*Picture from https://www.journal-topics.com/articles/gang-graffiti-strikes-multiple-des-plaines-homes/

TECHNOLOGIES

- **PYTHON – preferred language for machine learning**
- **GOOGLE COLAB – Google platform for handling sets of data**
- **LIBRARIES:**

NumPy	**works with large tables of data**
Pandas	**adds more functionality to NumPy**
MatPlotLib	**plots large data sets and builds on NumPy**
Seaborn	**adds more functionality to MatPlotLib**

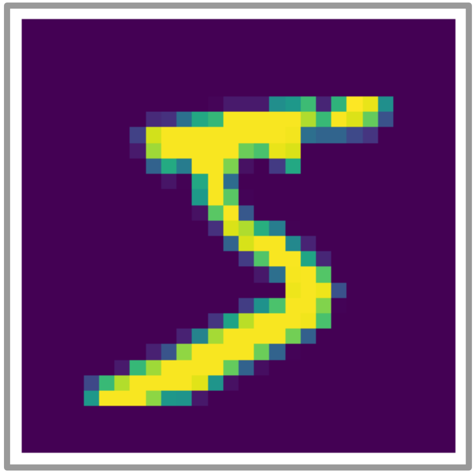
DATA SET

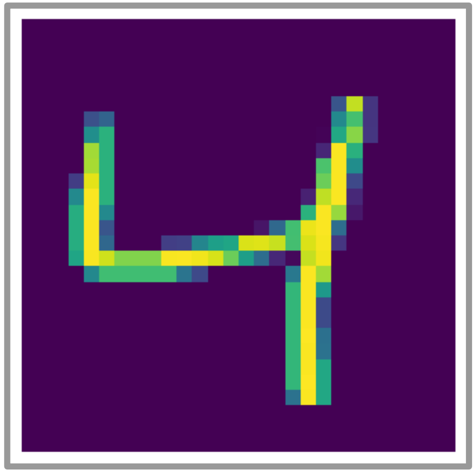
- **DATA SET USED TO TRAIN THE MODEL – MNIST**
- **MNIST - Modified National Institute of Standards and Technology**
- **IMAGES OF NUMBERS AND WHAT THOSE NUMBERS ARE**
- **DATA SET IS 784 COLUMNS AND 60,000 ROWS**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
1		index	labels	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
2	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	2	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	3	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	4	4	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	5	5	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	6	6	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	7	7	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DATA SET AND CLEANING

- EACH ROW IS AN IMAGE OF A HAND WRITTEN NUMBER





- DATA DOES NOT REQUIRE MUCH CLEANING
- ONLY NEED TO DROP FIRST TWO COLUMNS

DATA TESTING

- 3 MODELS: GAUSS NAIVE BAYES, GAUSS NON-NAIVE BAYES, K NEAREST NEIGHBORS (KNN)
- PARAMETERS TO TWEAK: EPSILON, SCALING THE DATA, NUMBER OF NEIGHBORS (KNN ONLY)
- ONLY NEED TO CHANGE EPSILON FOR MAXIMUM ACCURACY
- THE STANDARD FOR MODELS TO BE CONSIDERED SUCCESSFUL: 87%

ACCURACY

RESULTS OF EACH MODEL

GAUSS NAIVE BAYES	80.2%
GAUSS NON-NAIVE BAYES	95.8%
KNN	TOO LONG!

RECAP

- WE WANT TO PROCESS GRAFITTI IMAGES QUICKLY AND ACCURATELY
- DATA FOR TRAINING MODEL IS VERY ADEQUATE
- THE GAUSS NON-NAIVE BAYES MODEL CAN PROCESS THE IMAGES QUICKLY AND ACCURATELY – 95.8% !
- PROGRAM IS A SUCCESS AND SHOULD BE VERY HELPFUL FOR OFFICERS AND DETECTIVES ON THE STREET
- THE RESULTING DATA CAN BE STORED AND DISPLAYED AS REQUESTED BY THE DEPARTMENT

THANK YOU FOR YOUR TIME

THANK YOU FOR YOUR SERVICE

NAIVE VS NON-NAIVE

- **NAIVE IS SIMPLE AND ASSUMES NO RELATIONSHIP BETWEEN THE COLUMNS OF DATA.**
- **NON-NAIVE ASSUMES RELATIONSHIPS BETWEEN THE COLUMNS OF DATA.**

EPSILON

- **A number added into the calculations.**
- **Represents an arbitrarily small quantity, an error term, or a parameter related to model sensitivity to outliers.**
- **Needed to avoid division by 0.**
- **Calculated purely based on observation.**
- **Each calculation takes time and resources to complete.**

epsilon	accuracy
0.001	78.6%
0.1	83.7%
10	90.3%
100	93.7%
1,000	95.8%
5,000	95.7%
10,000	95.0%

SCALING THE DATA

- Changes the data into numbers from 0 to 1.
- Scales the data to a certain range which improves performance.
- Was not necessary for this data set.

NUMBER OF NEIGHGORS

- THE KNN (K-NEAREST NEIGHBORS) IS A MODELING TECHNIQUE WHERE THE DISTANCE OF EACH PIECE OF DATA FROM EVERY OTHER PIECE OF DATA IS CALCULATED.
- NOT EVERY PIECE OF DATA IS CALCULATED, ONLY A CERTAIN AMOUNT WHICH WE CALL ‘K’.
- EXTREMELY RESOURCE HEAVY.